# PhaseEval: expert-graded phase identification for AI agents

OSI LLM Hackathon, MIT, October 21 to 22, 2026. Team: Detrix (Gabe Graves). Video linked from the social post. Code: https://github.com/gabegraves/phase_eval. PhaseEval is the expert review, triage, scorecard and qualification product; AgentXRD is the analysis agent and harness it reviews.

## The problem

What AgentXRD does: it is our XRD analysis agent. Given a powder pattern and the sample's declared precursors, a language model (here Qwen3.6-27B on our own GPU) searches a crystal structure library for candidate phases, runs Rietveld refinements in BGMN through recorded tool calls, and reports the phases, their amounts and the fit it relied on. PhaseEval is where an expert checks that answer.

Checking an agent's diffraction answers takes expert time. A low fit statistic does not prove the phase list, so someone who knows diffraction has to look. PhaseEval shows each answer's checks, routes strong flags to an expert, supports blind judgment by unexposed raters, and turns eligible stored judgments into test cases that later models are graded against.

## What we built

A triage policy reads each answer's checks (a residual peak at or above 5 percent of the strongest net peak, an answer that differs from its cited fit) and routes the answer to an expert; weaker peaks are listed for review. Routing runs on the served build; the threshold is provisional. The expert sees the evidence first: the fit with unexplained peaks flagged, each phase's calculated peaks, and chemistry evidence per phase (charge balance, hull distance with its source, element balance). Then the agent's write-up, with record-backed checks on selected claims, and the rubric: is each phase supported by the pattern, chemically plausible for this sample, are the amounts defensible on their stated basis, does the fit support the conclusion. For unexposed raters the reference answer is withheld until the server has stored the sign-off. Analyses from GSAS-II (.gpx, opened only in a sandboxed subprocess) and BGMN/Profex bundles import into the same review path. An expert can mark any refinement step, and can act: add or remove a phase and refit on BGMN, creating a child attempt beside its parent. Agents use the same recorded action path. Eligible human judgments become qualification cases under the configured review rules (consent, quorum, exposure; unresolved judgments stay out): required phases from the accepted analysis, rejected analyses, free phases where raters disagree, and amounts only when the accepted fit is quantitative, judged defensible, and carries a declared tolerance. A model's answers are graded against those cases with accepted and rejected rates kept apart. Two model judges, GPT-6 Astra and Claude Opus, isolated and tool-less, each ran once over the 18 stored answers (valid labels on 12 and 16 of 18), stored apart from expert labels. They agree on 81 of 86 comparable items, but Opus said yes on 146 of 148, so kappa is 0.42. No judge-versus-expert number exists, because no expert labels exist. A scorecard lists every agent run by workflow step, graded against the published reference, deterministic checks and the two model judges (no expert labels exist yet), and keeps demonstration pairs apart from anything eligible for review or training. Eligible records (episodes, answer snapshots, labels, step feedback) export as versioned JSON with published schemas and a validator, following the process Periodic Labs published.

## The case in the video

DEV006 is a public DARA benchmark pattern (Zenodo 10.5281/zenodo.17410051, CC BY 4.0): the eight-minute scan of an unheated mix of NiO, Li2CO3 and Bi2O3. The video shows an episode by Qwen3.6-27B running locally: the three precursors and a residual peak near 24 degrees at about 3 percent of the strongest net peak, where the precursors contribute almost nothing. The check lists it as a weak peak for review, below the provisional K2 flag; this case was selected for the audit sample. The owner has seen the demo reference, so the labels in the video are revealed demonstration labels, not blind, and they are excluded from evaluation. The published reference lists only the three precursors. DARA's own first solution had reported Bi2CO5, and a second model, GPT-6 Astra, added it at about 2.7 percent of modeled crystalline mass. Reference-match checks and model-judge opinions do not adjudicate whether Bi2CO5 is present. Whether Bi2CO5 is present remains unresolved; the owner demonstration shows the review and refit tools, not expert adjudication.

## Prior work, event work, and what we do not claim

Before the event: the BGMN backend, the .xrdml reader, the agent tool routes and the harness. During the event: the recorded agent episodes, the attempt tree, the review workbench with rubric labels and step feedback, chemistry evidence, triage, qualification, GSAS-II and BGMN/Profex import, the export with schemas, the two-model judge runs, the evaluation dashboard, and property-based regression checks that no reference answer or other rater's label reaches a blind rater's view on the covered routes (one served review path was found open and fixed during the event). Qualification is shown in tests on synthetic fixtures; triage runs on the served build with a provisional threshold. PRIOR_WORK.md lists the split; CLAIMS.md traces every number. No model was trained; we claim no learning gains, held-out accuracy or routing recall, and three episodes per model per case is no accuracy comparison.

## Why it matters

Expert time is the scarce input. This spends it where it counts and keeps eligible judgments as test cases. One rubric, one export format and a judge ensemble that can be qualified against held-out expert judgments let it scale past one lab, and the method applies to any analysis where an expert can say what the data supports.